

AGORA

Supplementary Material

One Sentence, One Living World

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This supplement accompanies *Agora: One Sentence, One Living World*. It contains reproducibility details, complete construction and execution audits, planner-input and action-space diagnostics, historical records, the returned visual-survey protocol and counts, internal procedure controls, a prospective memory-intervention design, and case studies. The intervention is planned work; it is not counted as a completed result. Sections, figures, tables, equations, and pages use an S prefix. References to the main paper are explicitly identified; bibliography numbers are local to each document.

S1 Reproducibility and Evidence Audit

Generation settings and aggregation. The frozen protocol uses `gemini-3-flash-preview`, temperature 0.2, and low thinking level. Specialized stages have an 8,192-token output cap; the monolithic call has a 24,576-token cap. Five primary premises have three repetitions each and five held-out premises have one each. A primary premise passes a binary completion endpoint when a majority of its repetitions pass; the five held-out premises use their single observed outcome. Costs summarize all trials within each split. Failed outputs are retained. Quality-evaluator definitions and the held-out set were frozen before held-out generation.

The manuscript’s construction figures read the frozen file `paper_evidence_summary_20260729.json`. The six quality proxies are merchant-inventory pass rate, premise-item recall, premise-agent recall, role uniqueness, room visual-canon adherence, and wardrobe-canon term coverage. They are reported as proxies and are not substituted for human judgments. The seeded validator report is included alongside these summaries.

Accepted interaction runs. The accepted records are `mw3_aurora`, `mw3_clockwork`, `mw3_mycelium`, `mw3_sunken`, and `mw3_tidal`, collected on August 4, 2026. They use Gemini 2.5 Flash through AI Studio, two rounds, activation probability 0.12, and seeds 26080411–26080415 in that order. Compiled profiles are reused, and image/video generation is disabled. The archived run configuration does not by itself establish an immutable provider model snapshot; decoding details absent from that record are not inferred.

Earlier `mw_` attempts are excluded because a shared mutable scenario directory caused cross-world contamination. Earlier `mw2_` attempts made no model progress because DNS was unavailable. The accepted runs place mutable scenario state in separate directories and pass the world-identity audit. These exclusions concern execution validity and are distinct from excluding an unfavorable behavioral outcome.

Success-count correction. Table S1 distinguishes action results from story events. The frozen metrics already reported 0.875; summing the raw round summaries recovers 49 successes from 56 results. This revision corrects the prose denominator without changing the recorded success rate or event count. The accompanying audit file includes the source paths, SHA-256 hashes, and extracted round counts so that this cor-

World	Successes	Results	Rate
Aurora	9	9	100.0%
Clockwork	11	13	84.6%
Mycelium	8	10	80.0%
Sunken	11	11	100.0%
Tidal	10	13	76.9%
Total	49	56	87.5%

Table S1. Audited action-result counts, summed directly from the accepted story bundles’ round summaries. The 61-result denominator in the previous manuscript was a transcription error.

rection is traceable.

Historical analysis. The pilot records are the guild run of April 22, 2026, the black-market run of April 19, and the cruise run of April 18. They are analyzed independently of the August interaction runs and the planned intervention. Observed-ID counts are 103, 99, and 100, respectively; these are counts of recorded actor/target identifiers, not a claim that all runs had an identical realized population.

Artifact scope. The release contains the paper sources, BibTeX database, figure programs, frozen figure inputs, evidence-audit summaries, and the English visual survey returns with their fixed allocation, extracted choices, completed Excel workbook, and descriptive-analysis script. The bundled inputs reproduce the figures and reported summary arithmetic; full runtime replication additionally requires the original implementation, configuration, model access, and raw execution records. The expanded release reanalyzes existing August 25 generation and execution records. This revision incorporates eleven returned survey files, ten eligible for analysis. No new model call or paired memory-intervention result is claimed. The figure and table inputs retain the response exclusions and every Not sure choice, so the published summaries can be regenerated.

S2 Extended Evidence: Counts and Harness Audit

Source selection. Generation outcomes come from the 24 result records under the archive’s `confirmatory_runs` directory. A first-pass completion is a complete artifact with no recorded localized generation retry. Retry lists for some failed

Recorded model	Complete	First pass	Executed	Inventory/location	Approved	Non-speech
Gemini 2.5 Flash	3/3	2/3	174/216	50/174	45	34/45
Gemini 3.1 Flash Lite	3/3	2/3	185/216	75/185	89	50/89
Gemini 3.1 Pro Preview	3/3	0/3	190/216	67/190	19	13/19
Gemini 3.5 Flash Lite	1/3	1/3	171/216	42/171	102	45/102
Gemini 3.7 Flash	3/3	3/3	216/216	19/216	22	4/22
GPT-5.6 Luna	0/3	0/3	N/E	N/E	N/E	N/E
GPT-5.6 Sol	3/3	1/3	215/216	87/215	19	4/19
GPT-5.6 Terra	1/3	0/3	214/216	58/214	18	0/18
Total	17/24	9/24	1365/1512	398/1365	314	150/314

Table S2. Retrospective counts without reviewer scores or rank ordering. Construction columns use three prompts per model. Execution columns use only the first prompt’s generated artifact and three repeated trajectories. N/E means no eligible artifact; it is not a runtime zero or a human preference. “Non-speech” includes status effects and is counted from approved proposals.

cases were not preserved in the original format; we therefore do not estimate total repair incidence from those lists. Failed cases remain failed. The 21 execution sources are the corrected `runtime_trajectories_v2/replicate_*` records. Earlier runs affected by human placement and transport defects are not substituted or pooled. All raw result and trajectory records used here are frozen with SHA-256 hashes and recomputed by the bundled audit script.

Input scope of the earlier construction comparison. The ten requests in main-paper Section 5 are *world briefs*: some include several sentences specifying institutions, cast size, room requirements, and visual style. They do not test strictly one-sentence input. For the comparative questionnaire, the archived single-pass prompt contains an original-request JSON exactly matching the corresponding compositional request for all ten selected first replicates. This confirms a shared user request; the method-specific system instructions and total computation differ. The three later prompts are separate, genuinely one-sentence semantic inputs.

Execution schedule and observability. Each four-round block is planned from one snapshot; intermediate actions can make later planned actions invalid. At each checkpoint the policy receives room connectivity, all-agent summaries, the action catalog, up to four open action examples, the current intervention, and the 24 most recent events. Decoding uses the recorded temperature 0.75 and low thinking setting. There is no archived claim of matched random seeds across providers. A model generates and controls its own world, so world difficulty, generation quality, and policy quality are confounded. The harness is an executable-plan stress test, not an information-asymmetric social experiment.

Scripted human and recovery probes. The round-5 request and round-21 follow-up are supplied by the harness. When needed, the harness moves the human into an occupied room before a request or rejection probe; this movement is not counted among model actions. The round-9 probe requests transfer of an item the human does not possess. All 21 such probes are rejected without a state-hash change. A resource shock at round 13 and a cross-room instruction at round 17 are likewise scripted. Target acceptance in model-generated proposals is a declared field, not an independently elicited human

response.

What counts as a consequence. The inventory/location endpoint includes successful movement as well as inventory effects, and therefore is not a measure of economic transformation alone. The non-speech endpoint includes temporary statuses and describes the requested effects of approved proposals; approval does not establish that every intended semantic consequence occurred. The 147 failed model actions contain both before/after hashes and a false state-change flag. Five references to nonexistent event IDs occur in failed Gemini 2.5 Flash actions; no successful action in the archive has such an unknown reference. Thread labels and event references are supplied by the planner, so retaining them is not independent proof of causal coherence.

Statistics and reproducibility. The independent scope is 24 generation cells and 21 trajectory records, not 1,512 independent experimental units. We report counts and the observed range over three runs, without asymptotic confidence intervals, significance stars, or a composite leaderboard. The audit reads raw outcomes and mutations and does not load the identity-visible expert review or its weighted scores. The source snapshot also includes the planner harness, coordinator validation, and custom-action handler used to interpret the records. Source inspection supports the mechanism explanation but does not replace fresh execution tests against an immutable historical checkout.

S3 Planner Input and Action-Space Diagnostics

Question and experimental unit. This compact ablation study documents behavior under changes to recent event context and open-proposal access in the centralized execution harness described in main-paper Section 8. Its counters characterize the implemented planner and handlers; the main paper’s mechanism claim concerns how generated state constrains actions and retains their effects. Aggregate execution counts are not a measure of the mechanism’s novelty or overall value. It uses the existing Clockwork Rain Conservatory and Tidal Embassy worlds. Each world has one frozen initial agent state and one fixed set of rules, reused across three configurations and three stochastic repeats. Gemini 2.5 Flash is the only planning model. The resulting 18 trajectories are distinct from the earlier

cross-model audit and its model-specific generated worlds.

Conditions. Full supplies the current state, generated action catalog, current tasks, current scripted intervention, and the 24 most recent logged events. No event history sets only that last event list to empty. Current locations, inventories, status summaries, tasks, and the current intervention remain available. This is an input-history ablation, not removal of all historical information or a test of decentralized episodic memory. Catalog only keeps state, tasks, and history but removes the proposal option from the prompt and JSON schema; the execution boundary also rejects proposals. Catalog actions and adjacent-room movement remain available. As implemented in this harness, catalog actions update statuses and relationships; they do not transfer or create inventory. Its inventory limitation is therefore partly structural.

The earlier prospective Tidal configuration is not used as the manipulation: its zero-valued memory limits are clamped to at least one by the legacy helper, and it also changes task/cohort limits. Directly masking the planner’s event list avoids treating those settings as a clean memory ablation. Tests verify that all other supplied fields remain identical when constructing full and history-masked prompts from the same state.

Common computation and scheduling. Every trajectory has 24 rounds and six planning calls, at rounds 1, 5, 9, 13, 17, and 21. Each call requests 12 actions, three for each of four rounds. An AI character may act at most once in each round; the scripted human is a target. Calls use temperature 0.45, disabled thinking, a 6,144-token output cap, and a 90-second timeout. At most two API attempts, including retries for invalid JSON, are allowed. There is no alternate-model fallback. All conditions have the same call and output limits; removing input history reduces input tokens, so total token use is not equalized. Repeats are new stochastic API calls, not claims of a provider-controlled random seed.

The schedule includes the existing human request, rejected unowned transfer, resource shock, cross-room request, and human follow-up. Intervention rounds and rules are shared; their recipient can depend on the state reached within each trajectory. Current request records remain visible in every condition. No actual participant is interacting during this evaluation. A fixed local randomization orders the 18 run jobs, with up to four trajectories processed concurrently. There is no adaptive stopping based on condition performance.

Protocol and retained outputs. The implementation was checked using three separate two-block pilot runs on Clockwork. These six pilot calls are excluded. The study protocol, all input states, and the runtime source snapshot were then hashed before the 108 formal planning calls. This is a local pre-execution specification, not an external preregistration. Model identifiers describe the requested API configuration; an immutable provider snapshot is not claimed.

Every block preserves the supplied prompt payload and hash, JSON schema, returned plan, call telemetry, executed events, and resulting state. Failures are retained. The source snapshot contains the actual coordinator and action handlers used by the experiment. The replay verifier reconstructs interventions and re-executes every recorded action, checks the

Configuration	Success	Material	Inventory
Full	349	196	14
No event history	352	177	24
Catalog only	375	193	0

Table S3. Aggregate planner diagnostics on two fixed worlds. Each row has 432 scheduled action slots across six trajectories. Success counts executed actions; Material counts actual inventory or location changes among successes; Inventory is its inventory-changing subset. Status and relationship changes alone do not count as material.

resulting state and event hashes, and independently recomputes the reported counts without further LLM calls.

Metrics and interpretation. The primary denominators are the 72 scheduled action slots per trajectory, including slots lost to a planning failure. We report successful executions and successful actions that change inventory or location. The latter requires an actual difference in the sorted per-agent inventory/location snapshot; relationship changes, status updates, and descriptions of intended effects alone do not count. Inventory and movement counts are shown separately. A location change does not by itself establish useful task progress. We also retain proposal counts, complete-block counts, failed-state changes, and provider-reported token usage. No weighted overall-quality score is formed.

Comparisons are paired within world, model, and repeat index. Repeats share the same two worlds, and actions within a trajectory are dependent. We report raw counts and descriptive differences, without treating the 1,296 action slots as independent samples or claiming a population effect. The study does not establish decentralized social coordination, human enjoyment, or a universal benefit from longer history.

Observed execution and cost. Table S3 retains aggregate counts for every condition. Full, No event history, and Catalog only execute 349, 352, and 375 of their 432 scheduled actions, respectively. Their inventory/location-changing successes number 196, 177, and 193; inventory-changing subsets number 14, 24, and zero. The catalog-only material changes are all moves, consistent with its restricted handlers. These totals do not establish a benefit from supplied event history or measure successful completion of the scripted tasks.

All 108 formal planning blocks returned valid 12-action plans. One truncated response required a second attempt, giving 109 API attempts. The 1,296 planned slots produced 1,076 successful actions and 220 failures; every failure leaves the recorded state unchanged. Of 566 inventory/location-changing successes, 528 are moves and 38 change inventory. Full executes 30 of 57 proposals; No event history executes 93 of 138. These counts do not measure task utility.

Provider telemetry records 1,821,701 input tokens and 201,841 output tokens across formal attempts, including the truncated attempt. Mean total tokens per trajectory are 137,731 for Full, 74,598 for No event history, and 124,928 for Catalog only. Cache-token counts are retained separately as a subset of input usage, not added again. The six technical-pilot calls are outside these totals.

Relative to Full, the mean paired execution difference is

World	Configuration	Rep.	Success	Material	Inventory	Move	Proposals
Clockwork	Full	1	64/72	44	1	43	1
Clockwork	Full	2	57/72	32	4	28	4
Clockwork	Full	3	55/72	31	1	30	3
Clockwork	No event history	1	59/72	30	5	25	11
Clockwork	No event history	2	55/72	38	8	30	9
Clockwork	No event history	3	63/72	29	8	21	22
Clockwork	Catalog only	1	68/72	43	0	43	0
Clockwork	Catalog only	2	68/72	27	0	27	0
Clockwork	Catalog only	3	60/72	30	0	30	0
Tidal	Full	1	67/72	37	6	31	6
Tidal	Full	2	58/72	24	2	22	2
Tidal	Full	3	48/72	28	0	28	14
Tidal	No event history	1	56/72	20	3	17	26
Tidal	No event history	2	58/72	32	0	32	19
Tidal	No event history	3	61/72	28	0	28	6
Tidal	Catalog only	1	65/72	31	0	31	0
Tidal	Catalog only	2	53/72	33	0	33	0
Tidal	Catalog only	3	61/72	29	0	29	0

Table S4. All 18 formal trajectories. Each trajectory has 72 planned action slots. Material counts successful actions with actual inventory or location changes; Inventory and Move separate these consequences. Proposals counts successfully executed proposals. Repeats share a frozen world and are not independent world samples. Pilot runs are excluded.

+0.7 percentage points for No event history, with repeat differences from -15.3 to $+18.1$; its material-change difference is -4.4 points, ranging from -23.6 to $+11.1$. Catalog only has a mean execution difference of $+6.0$ points (range -6.9 to $+18.1$) and a material-change difference of -0.7 points (range -8.3 to $+12.5$). These are descriptive ranges over six paired repeats on two worlds, not confidence intervals. The variation and handler constraints limit the interpretation to these recorded trajectories and effect types.

Trace provenance for the main-paper example. The example in main-paper Section 8 comes from the Clockwork Full trajectory, repeat 1, event `model_r09_slot09` (actual round 11). It is the first inventory-changing success in the first lexicographically ordered Full trajectory with such an event, selected after collection to illustrate the mechanism. In `block_09.json`, Thistle holds one Corrosive Inkwell and Elara holds none. The event records an approved transfer of that item; `block_13.json` reverses those holdings in the supplied planner observation. The target-response field is unknown: this outbound gift illustrates transfer checks, not independently observed recipient acceptance. The same trajectory’s scripted event `coordinator_rejection_r09` rejects a transfer of an unowned item without changing state. These records are included under `llm_ablation/runs/`; the illustrative selection adds no model calls and does not replace the complete counts above.

S4 Exploratory Historical Trajectories

We separately analyze three prior runs configured around approximately 100 agents: an adventuring guild (25 rounds), an underground black-market auction (20), and a storm-struck luxury cruise (20). They predate the current hypotheses and

differ in configuration and social context. We reconstruct undirected actor–target networks, route entropy, and sums of directed relationship changes using the same metric definitions.

Between 23.4% and 36.5% of events revisit an existing dyad, while each world contains 229–282 unique dyads (Table S5). Route entropies are 2.889, 2.619, and 2.524 bits for guild, black market, and cruise, respectively. Because the repertoires differ, these values describe each trace’s variety; they are not directly comparable measures of social intelligence.

The largest components contain only 11–13 participants despite many observed IDs. Figure S1 shows dyads continuing to accumulate after component sizes stabilize. This is consistent with additional ties forming predominantly within local clusters. Spatial encounter and activation policies can generate such structure, so its social interpretation must remain conditional on those mechanisms.

In the black market, influence/fear (+1194) exceeds trust (+179) and affection (+9); the guild and cruise instead accumulate mostly trust and affection. The pattern is compatible with each scenario’s institutions, but prompts, action definitions, schedules, and run histories are confounded. It motivates a controlled question rather than answering one. In particular, warning diffusion and collective rescue need separate outcomes: receiving information need not produce action [1]. Supplementary Section S6 specifies a paired memory intervention to examine that distinction.

S5 Returned Visual Survey: Protocol and Complete Counts

Version and evidence. This model-focused presentation retains both originally collected arms and all eligibility rules; no responses or unfavorable internal-control outcomes are re-

World	Rounds	Events	IDs	Dyads	Repeat	Largest comp.	Dominant state changes
Guild	25	444	103	282	36.5%	11	Trust +3473; affection +1745
Black market	20	299	99	229	23.4%	13	Influence/fear +1194
Storm cruise	20	309	100	235	23.9%	13	Trust +2576; affection +961

Table S5. Exploratory statistics from 1,052 historical events. IDs are distinct logged actor/target identifiers, not a fixed population denominator; the guild record contains 103. Relationship totals are simulated state changes.

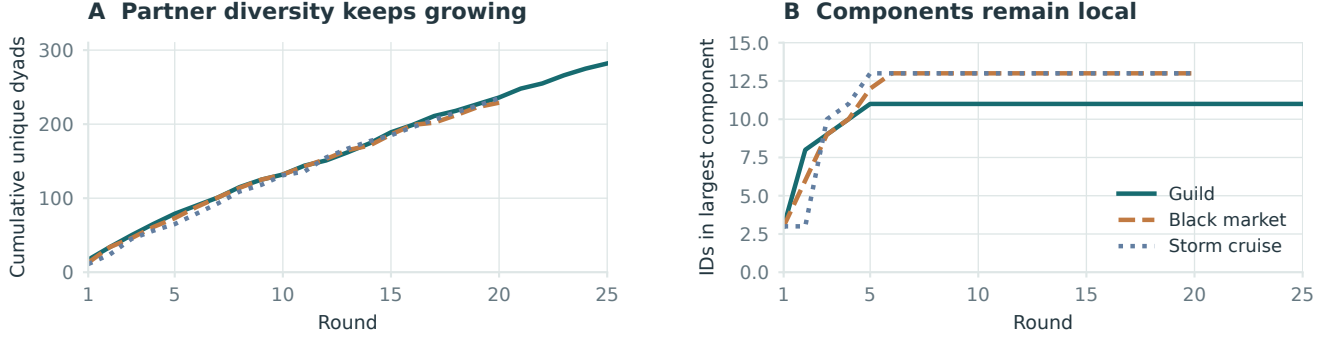


Figure S1. Longer traces distinguish activity from connectivity. Unique dyads continue to accumulate, while the largest observed components plateau early. These historical runs motivate testing transmission across local clusters; they do not identify its causal mechanism.

moved. The per-model pooled summary was added after collection. This analysis uses the seven-page, all-English visual packets V01–V11, released September 22, 2026, and the returned archive supplied on GitHub before this revision. It supersedes the text-based prospective evaluation in earlier manuscripts; older Q/S/P instruments are not pooled with these responses. Each coded return has six comparisons and eighteen answers. The source PDF form values are checked against the selected widget appearance states before decoding through the frozen allocation. No answers are inferred from a model, imputed, or edited. The filled Excel file preserves every return, with V04 marked for cover verification and omitted from all preference summaries.

Visual sampling. The image arm uses the one archived premise with complete matching render assets: a cliffside city that taxes spoken promises by weight. Each of seven world configurations contributes its unaltered map and the first idle-down frame of its first two requested characters, under the same framing rule. These are finalized pipeline assets, including recorded repair steps, not independent fresh generations. The unavailable Luna artifact is not replaced or counted as a preference loss. Terra’s unavailable third publishable character is not shown; no condition is judged on a third character. This instrument does not evaluate asset completeness, animation, or navigation.

Structure sampling and presentation. The ten matched requests use the fixed first archived replicate of full Agora and single-pass generation. The diagrams include all source rooms in source order, count inhabitants by their recorded home location, and show the count of large scene components with the first component label as an example. Lines denote containment, not adjacency. The same short English scene description appears on both sides. The comparison retains readable source records even when an objective completion contract fails. There are no matched baseline renderings: no un-

related or synthetic image is substituted. The conditions retain unequal total generation budgets.

Allocation and masking. Seed 20260922 fixes the schedule before response analysis. Every participant sees four distinct image pairs and two distinct structure pairs. The original plan has 44 image and 22 structure comparisons; each exact pair occurs twice, with one seed-selected pair per arm occurring four times. Every pair has balanced A/B orientation, and Agora appears on A once and B once for each participant. Excluding V04 removes four image and two structure comparisons, leaving the coverage counts reported in main-paper Section 6. Individual pair orientation is consequently no longer universally balanced. The twenty retained structure comparisons remain balanced overall: Agora is shown on each side ten times. The questionnaire hides source names, but visible artifacts can still reveal stylistic clues; analyst blinding is not claimed.

Questions and estimands. Q1 in both arms asks, “Which world would you rather explore?” Image Q2 asks which map is easiest to read; Q3 asks which characters fit their world better. Structure Q2 asks which world is easiest to understand; Q3 asks which places fit the scene better. All offer A, B, Same, and Not sure. Exploration measures *stated interest from the supplied evidence*, not enjoyment after playing. The diagnostic items measure different constructs across the arms and are never merged. This explicit separation follows the reporting concerns of Howcroft et al. [2].

Let $w_{jq}, t_{jq}, \ell_{jq}$ denote Agora wins, ties, and baseline wins for request j and item q , after eligibility filtering. With abstentions excluded,

$$\hat{p}_{jq} = \frac{w_{jq} + t_{jq}/2}{w_{jq} + t_{jq} + \ell_{jq}}, \quad \hat{p}_q = \frac{1}{10} \sum_{j=1}^{10} \hat{p}_{jq}. \quad (\text{S1})$$

The ten-request mean is shown only if all ten denominators are nonzero. All three items meet this condition. Pooled vote

scores, provided only as a weighting diagnostic, are 35.0%, 42.5%, and 36.8%, versus equal-request scores of 40.0%, 32.5%, and 37.5%. Clockwork has four votes and is preferred for clarity but not exploration; reporting the estimator explicitly prevents this imbalance from silently changing the result. Image scores use the same tie and abstention rule separately within each exact model pair, with no cross-prompt or global model ranking.

Eligibility, consent, and missing administration records. Ten covers affirm age 18 or older, English comprehension, voluntary participation, and no prior work on the system or exposure to detailed results. V04 has eighteen choices but no cover answers; blank checkboxes are treated as unconfirmed, not as an affirmative response or a known refusal. The exclusion rule was included in the original release and does not depend on preferred source. The archive supplies no names or demographics beyond the cover confirmations, nor independent verification that codes correspond to distinct people. The organizer reports online volunteer recruitment with no compensation, and identifies V04 as an accidentally omitted cover. This explanation does not supply the missing participant confirmations. No sampling frame or probability-sampling procedure is documented; recruitment is treated as an online volunteer sample. Actual completion time and an ethics approval or exemption record were not supplied. No approval, exemption, representative sampling, or measured 10–15-minute completion time is claimed.

Response dependence and presentation checks. The 180 item responses come from ten coded returns; they are not 180 independent people. All image comparisons reuse seven artifacts on one scene, and all three questions reuse each displayed pair. We report counts and request-level variation, without item-level significance tests, Elo scores, or confidence intervals that would imply unsupported population precision. Image Q1 selects A 15 times and B 25 times; structure Q1 selects A 11 times and B 9 times. This descriptive side asymmetry is retained for inspection, but does not identify a position effect separately from artifact preference. No respondent is excluded for a repeated choice pattern or for favoring the baseline. Very small per-pair counts make agreement coefficients unstable; the full counts are reported instead.

Image identity key. M1: Gemini 2.5 Flash; M2: Gemini 3.1 Flash Lite; M3: Gemini 3.1 Pro Preview; M4: Gemini 3.5 Flash Lite; M5: Gemini 3.7 Flash; M6: GPT-5.6 Sol; M7: GPT-5.6 Terra. These names preserve archived backend identifiers and do not independently verify product naming, an immutable provider snapshot, or common inference budgets. The actual visual artifacts, rather than the labels, are the measured objects.

S5.1 Internal Generation Procedure Control

This supplementary comparison varies the generation procedure within the Agora implementation. Both conditions use its world representation and rendering conventions; they are not independent world-generation systems. The total generation budgets also differ, so this is a procedure control, not a compute-matched component ablation. The returned results

below are retained in full and must not be interpreted as Agora’s performance against an external system. For each item, a preference for Agora scores 1, a tie 0.5, and a baseline preference 0; “Not sure” is excluded from that item’s preference denominator and reported separately. We average within each request and then equally across all ten requests, as specified in the released workbook. This prevents the request receiving four ratings from dominating the overall estimate.

Respondents select the baseline in thirteen of twenty exploration choices, versus seven for Agora. Agora’s equal-request preference is 40.0% for exploration, 32.5% for world clarity, and 37.5% for place–scene fit (Table S8). These observations favor the single-pass condition on the displayed containment diagrams. They neither benchmark Agora against a separate system nor measure the contribution of its shared representation, renderer, or runtime. A causal claim about decomposition would additionally require equalized budgets and controlled retries.

The pattern varies by request (Figure S2). Both Tidal raters prefer Agora for exploration. All four Clockwork raters prefer the baseline for exploration while choosing Agora for clarity. Understandability and desire to explore are thus distinguishable judgments in these returns; we do not sum them into a single quality score. One or two raters on most requests cannot identify why a design was preferred, and the diagram omits institutions, social behavior, and actual play. Equal-compute generation and interactive evaluation remain separate open tests.

S6 Prospective Tidal Memory Intervention

Status. This supplementary section describes planned work, not an additional completed experiment. The project contains a local protocol and a one-round full-memory smoke run; it does not contain completed paired 32-round results supporting a causal memory claim. No external preregistration identifier is available in the release. The changes to analysis and measurement specified below must be frozen before confirmatory runs and are not represented as an already registered plan.

S6.1 Intervention and Hypothesis

The 25-agent Tidal Embassy world is initialized once and reused. At round zero, three agents in the lower archive privately learn that a flood threatens the only surviving peace treaty unless it is moved and authenticated. The full-memory condition retains prior social episodes. The ablation removes prior-round social episodes from decision context while preserving current locations, inventories, public facts, task state, and directed relationships. Thus it tests access to episodic social history, not removal of all memory: tasks, current knowledge, and relationships can also carry historical signals.

The planned seeds are 26080311, 26080312, and 26080313, with 32 rounds per condition. Initial profiles, world configuration, model and decoding settings, and activation schedule are matched within each pair. Activation draws should be precomputed with a separate random stream; reusing a seed alone does not guarantee matching if the conditions consume randomness differently. API nondeterminism remains a source of variation.

The hypothesis is that access to earlier social episodes increases warning diffusion and coordinated rescue within this

Request	Explore	Clarity	Scene fit
Archive Of Borrowed Gravity	50% (1/0/1/0)	25% (0/1/1/0)	50% (1/0/1/0)
Aurora Court Of Migrating Cities	50% (1/0/1/0)	100% (2/0/0/0)	100% (2/0/0/0)
Cartographer Lung Exchange	0% (0/0/1/0)	0% (0/0/1/0)	0% (0/0/1/0)
Clockwork Rain Conservatory	0% (0/0/4/0)	100% (4/0/0/0)	25% (1/0/3/0)
Intertidal Embassy For Extinct Rivers	0% (0/0/2/0)	0% (0/0/2/0)	0% (0/0/2/0)
Museum Of Future Debts	0% (0/0/2/0)	0% (0/0/2/0)	0% (0/0/2/0)
Mycelium Patent Bazaar	100% (1/0/0/0)	0% (0/0/1/0)	0% (0/0/1/0)
Night Market Of Unfinished Weather	50% (1/0/1/0)	50% (1/0/1/0)	50% (1/0/1/0)
Sunken Satellite Monastery	50% (1/0/1/0)	0% (0/0/2/0)	50% (1/0/1/0)
Tidal Embassy Of Lost Languages	100% (2/0/0/0)	50% (1/0/1/0)	100% (1/0/0/1)

Table S6. Every structure request. Each cell gives the Agora preference score and counts (Agora wins / ties / baseline wins / Not sure). The requests, not the number of ratings, are equally weighted in the main result.

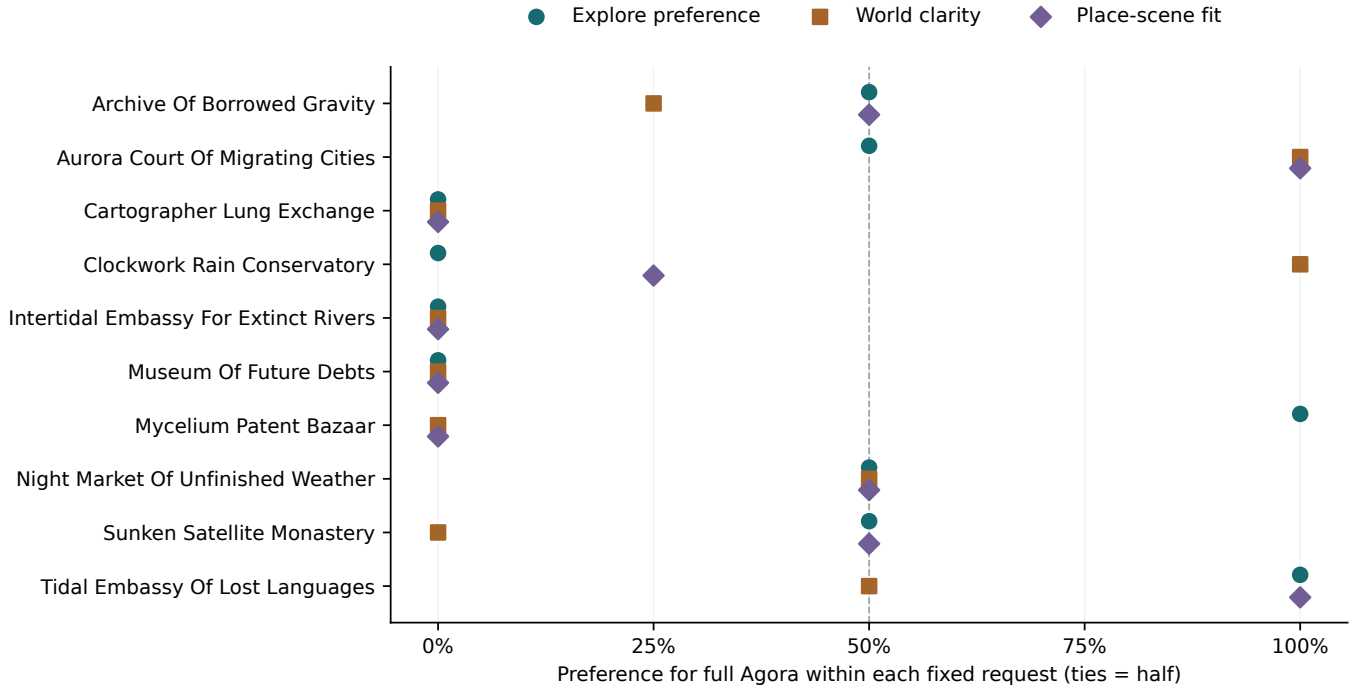


Figure S2. World structure: preferences vary by request and item. Each point is the fraction favoring full Agora for one request and question, with a half vote for a tie. The analysis includes 20 comparisons: seven requests have two raters, two have one, and Clockwork has four. For Tidal scene fit, one of two raters selects Not sure. These are descriptive fixed- artifact scores; the dashed line marks equal preference, not a significance threshold.

generated setting. Generative Agents motivates studying memory in community behavior [3]; the separation of information reach and collective action is also informed by contagion research [1].

S6.2 Outcomes and Measurement Controls

The primary endpoint is warning reach by round 16 among the 22 initially uninformed agents. Let $Z_i(16)$ indicate that the agent has expressed or recorded a prespecified treaty-threat proposition by that round. Then

$$D_{16} = \frac{1}{22} \sum_{i \notin I_0} Z_i(16), \quad |I_0| = 3. \quad (\text{S2})$$

Measurement must use an append-only audit stream outside the ablated decision memory; otherwise deleting memory could mechanically reduce measured reach even after successful transmission. The audit record is not fed back to agents. Initial warning availability must be preserved equally through its first intended use, so the ablation does not inadvertently remove the treatment trigger itself.

Secondary outcomes are first cross-room transmission time, rescue-task participation and role diversity, treaty relocation and authentication, unique and repeated dyads, largest component, object transfers, relationship changes, and rule violations. Unreached transmission or rescue events are reported as censored at the horizon. Two condition-blind annotators should label candidate episodes under a frozen proposition codebook;

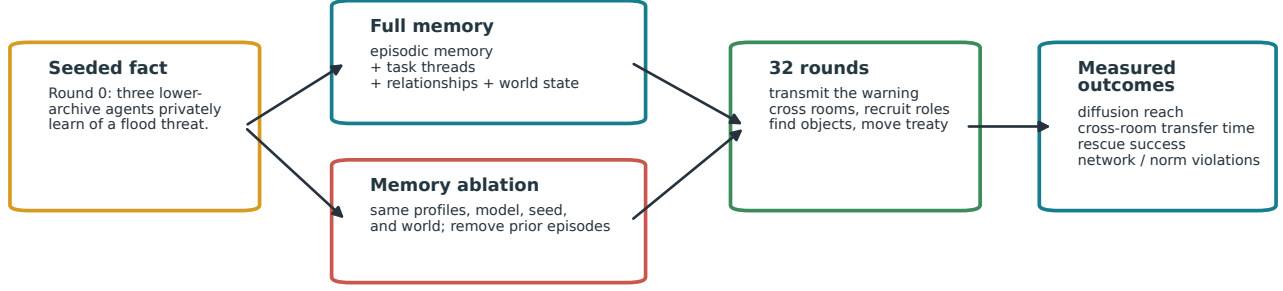


Figure S3. Prospective design; no paired results reported. Matched initial states receive the same private warning. Conditions differ in access to prior social episodes. Diffusion and material rescue are distinct outcomes. The planned unit of comparison is a seed pair.

Pair	Explore	Map	Characters
M1–M2	2/0/0/0	2/0/0/0	1/0/0/1
M1–M3	1/0/0/0	0/0/1/0	1/0/0/0
M1–M4	2/0/0/0	2/0/0/0	2/0/0/0
M1–M5	1/0/0/0	1/0/0/0	1/0/0/0
M1–M6	1/0/1/0	1/0/1/0	1/0/1/0
M1–M7	1/0/1/0	1/0/1/0	1/0/1/0
M2–M3	1/0/1/0	0/1/1/0	1/0/1/0
M2–M4	2/0/0/0	0/0/2/0	1/0/1/0
M2–M5	2/0/0/0	0/0/2/0	1/0/1/0
M2–M6	1/0/0/0	0/0/1/0	1/0/0/0
M2–M7	0/0/1/0	0/0/1/0	0/0/1/0
M3–M4	1/0/1/0	2/0/0/0	1/1/0/0
M3–M5	0/0/2/0	0/0/2/0	1/0/1/0
M3–M6	0/0/2/0	0/0/2/0	0/0/2/0
M3–M7	0/0/2/0	0/0/2/0	0/0/1/1
M4–M5	1/0/1/0	0/0/2/0	1/0/1/0
M4–M6	2/0/0/0	0/0/2/0	2/0/0/0
M4–M7	2/0/2/0	0/0/4/0	2/0/2/0
M5–M6	2/0/0/0	0/0/2/0	2/0/0/0
M5–M7	0/0/2/0	0/0/2/0	0/0/2/0
M6–M7	1/0/1/0	1/0/1/0	2/0/0/0

Table S7. Every image pair. Entries are first-source wins / ties / second-source wins / Not sure. Each row describes the same pair under three different questions; these are not independent respondent samples.

agreement and resolved disagreements must be reported before condition-level interpretation.

Manipulation checks verify that only the designated agents initially receive the warning, that paired initial states and activation schedules match, and that the ablation removes the intended history without exposing hidden facts through summaries or coordinator feedback. These checks implement the information-boundary concern highlighted by Zhou et al. [4].

S6.3 Analysis and Precision

The independent comparison unit is a seed pair, not an event or an agent. For seed s , report $\Delta_s = D_{16,s}^{\text{full}} - D_{16,s}^{\text{ablation}}$, both condition trajectories, and the material rescue outcome. Report all

Item	Full	Base	Tie	?	Score
Explore	7	13	0	0	40.0%
Clarity	8	11	1	0	32.5%
Scene fit	7	12	0	1	37.5%

Table S8. Structure judgments from ten eligible returns. Full and Base denote Agora and the single-pass baseline, irrespective of randomized A/B labels; ? = Not sure. Counts pool twenty comparisons per item. Score is Agora’s *equal-request* mean, not a pooled vote fraction. No population superiority is inferred from these small counts.

three differences and their mean and range. Three pairs are too few for a stable bootstrap confidence interval or a well-powered confirmatory test; additional seeds and a prospectively justified sample size are needed for precise inference.

Qualitative selection rules are the earliest cross-room transmission, first multi-role rescue, largest trust reversal, and first failed coordination. Selections should be applied identically to both conditions and report when an event does not occur. The existing one-round smoke run is execution validation only and must remain outside these comparisons.

S7 Clockwork Episode

The diagram follows the short episode described in main-paper Section 7. It is an illustrative trace from one two-round run, not an independently replicated behavioral effect. Six actions form the central three-character sequence; the seventh concerns the market impact of the malfunction.

S8 Tidal Embassy: A Generation and Play Trace

We now follow *Tidal Embassy of Lost Languages* from premise to play. Its one-sentence premise describes a floating embassy where linguists, smugglers, ambassadors, and memory divers trade endangered languages as political assets. The example preserves intermediate images and refinement attempts, allowing each generative stage to be connected to the final playable world.

Step 1: Premise analysis and planning. The planning stage identifies the nouns that become state—embassy, languages,



Subset totals above; the full 12-event run has +3 trust and +21 influence/fear.

Figure S4. An audited two-round episode in Clockwork Conservatory. Six open actions form the central Marius–Valerius–Seraphina sequence; Lyra adds a seventh proposal about market effects. The seven-proposal subset has trust –29, affection –5, and influence/fear +17; the full run has different totals because it includes five other events.

translation rights, archives, smugglers, diplomats, and tides—and the verbs that imply gameplay—trade, translate, forge, authenticate, rescue, and reveal. It then proposes an institutional loop: access to archives produces linguistic objects; translators and forgers can translate or forge them; diplomats exchange rights and debts; customs officers can inspect transfers; flood events threaten stored knowledge. The resulting world plan connects entities, conflicts, resource cycles, and open questions.

Planning establishes what must exist and why it matters. Later stages determine spatial layout, visual appearance, and resource quantities. Missing activity loops are refined before visual generation; for example, translation rights must have a source, a use, and a means of transfer.

Step 2: Semantic structure. Semantic generation expands the plan into rooms, roles, items, institutions, knowledge, and conflict hooks with stable identities. In this world, the graph contains eight rooms, including the Phoneme Exchange, Mnemonic Garden, Embassy Customs and Quarantine, and Ballast Hold; 25 named main characters; and 20 distinct items. Explicit relations connect each character to a home, each possession to an owner, and each action to relevant role, knowledge, location, or resource conditions.

The generation process also preserves asymmetry. Shen Mo is a Phonetic Smuggler and Lexical Restorationist based in the Cartographer’s Nook. His inventory includes a translation ear-cuff, archive key, forged exchange ledger, dive helmet, ritual ink, and a fragment of the Unspoken Treaty. His private knowledge includes phonetic decay patterns and ventilation shafts linking Customs to the Mnemonic Garden. Another character receives different property, authority, and information. This uneven allocation is what later makes persuasion, trade, secrecy, and coalition formation possible.

Step 3: Visual direction and cross-modal identity. The visual direction converts semantic identity into a shared canon: Art Nouveau diplomatic architecture, floating platforms, brass infrastructure, bioluminescent language media, water intrusion, and a melancholic ritual tone. It also emits per-entity appearance anchors. Room prompts inherit the canon but add function-specific materials; character prompts inherit world palette and wardrobe rules but retain role silhouettes; props receive icons whose shapes communicate their use. Shared style

and entity-specific appearance jointly keep rooms, inhabitants, and objects recognizable as parts of the same world.

The visual direction guides each image independently while maintaining common architecture, materials, palette, and role silhouettes. A visual inconsistency can therefore be refined locally while the semantic identity of the world remains stable.

Step 4: Evolving world state. This stage determines what can change after the images are produced. It creates wallets, inventories, property libraries, knowledge assets, directed relationships, task threads, room occupancy, and world-event status. It also defines invariants. Item quantities remain nonnegative; physical transfers conserve quantity; private knowledge has an owner and an acquisition event; a character occupies one reachable coordinate; relationship changes are directed. The generated repertoire and open-proposal forms are linked to the same world state.

The evolving state connects generated descriptions through observable causes and consequences. The treaty fragment in Shen Mo’s inventory can later be inspected, given away, stolen through an allowed rule, or used as an input to another action. When ownership changes, the next observation shows the new owner.

Step 5: Integrated world representation. The integrated representation connects semantic, visual, spatial, and evolving state. It supports cross-domain questions: Does every home room exist? Does every main character have a visual asset and a walkable placement? Is an item used by an action defined and obtainable? Does a private fact mention an existing institution? Does a room’s purpose agree with its objects and activities? These dependencies guide local refinement.

If a character image is inconsistent, roles, rooms, and items remain fixed. If an item relation is missing, only the connected content is regenerated. Figure S6 illustrates this process: two room images are refined while the rest of Tidal Embassy remains unchanged.

Step 6: Visual generation. FLUX generates the map source, individual room floors, characters, and local objects from the shared visual direction. Character images are arranged into directional movement atlases, and room images are placed within the planned regions and connected by traversable corridors. Figure S5 shows the composed map alongside three source floors, three character atlases, and an object.

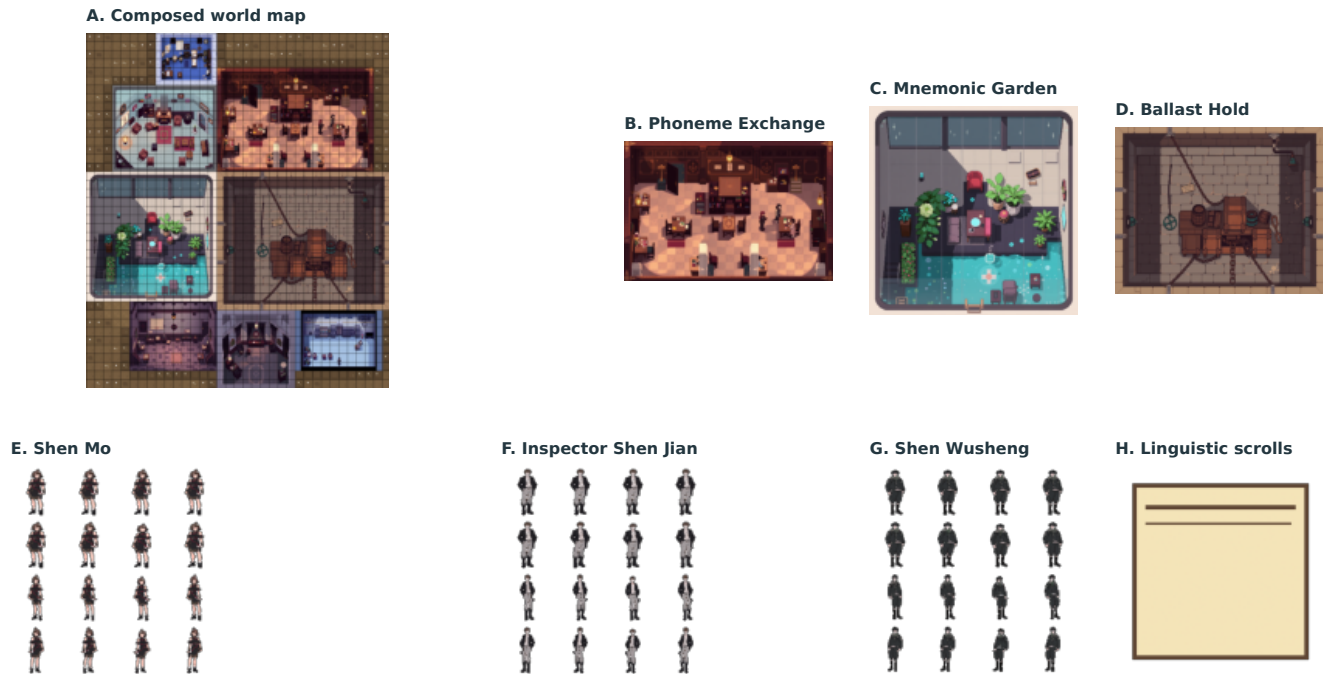


Figure S5. Generated visual elements from Tidal Embassy. The composed map contains eight generated floors (A–D show the map and three constituent rooms); E–G show the directional character atlases; H is a generated object. Shared visual motifs connect spatial, character, and object identity.

Functional visibility connects visual quality to use: room imagery must cover its playable area, characters must remain recognizable while moving, and furniture must preserve navigable paths.

Step 7: Visual and spatial evaluation. The evaluation examines semantic completeness, visual identity, traversable geometry, foreground coverage, directional character consistency, readable room composition, and visibility during play. The Diplomat’s Lounge required three real generation attempts. Attempt 1 was classified as a poster card with an edge blank-background ratio of 0.56. Attempt 2 removed the card appearance but retained a blank-margin failure with a maximum blank-row ratio of 0.73. Only the third, edge-to-edge playable floor was accepted.

In Agora, visual quality combines coherent identity, readable composition, and fitness for navigation and interaction. Preserved refinement attempts make the progression from attractive illustration to usable world space observable.

Step 8: Persistent play. The generated world begins from a fixed initial state and evolves independently for each session. Humans and AI agents encounter the same rooms, inhabitants, objects, actions, and consequences. Separate world identities keep concurrent multiworld experiments isolated.

Figure S7 connects character generation to embodied play. A crop of the interaction capture shows the selected agent. A complementary world view shows neighboring agents among generated furniture and room surfaces, while the directional atlas records appearance across locomotion. The interactive trial covers movement, messaging, item use, trade, continued play, and a model-mediated reply.

The 25-character set uses one canonical visual identity per

inhabitant. Front locomotion and side views preserve the same hair, clothing, body type, and palette, so an inhabitant remains recognizable while moving. Native 64×64 frames retain facial, clothing, and limb detail in the stored atlases.

Character evaluation covers single-person composition, silhouette readability, palette contrast, directional identity, absence of text and detached effects, and visibility against the room surface. Contact-sheet review additionally examines the complete cast together, revealing cross-view identity changes and visual outliers that are difficult to judge one frame at a time.

Successive review passes identified characters with problematic props, enclosing effects, orientation, text, detached sparkles, or insufficient contrast. These counts reflect different review stages rather than disjoint character sets. Localized regeneration produced a final set of 25 coherent, readable character atlases, all visible during the interactive trial.

Step 9: Interaction-informed refinement. During play, agents and humans observe bounded state and choose either a world-specific action or an open proposal. The World Coordinator evaluates effects and enacts successful transitions, as detailed in main-paper Section 3. Interaction traces also provide feedback upstream. An action that is never selected may be poorly named or irrelevant; an unreachable character suggests placement repair; an important object that never enters an exchange suggests a missing action; frequent open proposals around one intention suggest a useful addition to the visible repertoire. Observed life inside the world can therefore guide the next round of world refinement.

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Attempt 1: rejected

poster-card background
blank margin ratio: 0.56

Attempt 2: rejected

blank margin remains
max blank row: 0.73

Attempt 3: accepted

edge-to-edge playable floor
visual identity preserved

Figure S6. Localized visual refinement for the Diplomat's Lounge. The first image has a large poster-like margin; the second retains an extensive blank boundary; the third fills the playable floor while preserving the room's identity. Other rooms, characters, and world state remain unchanged.

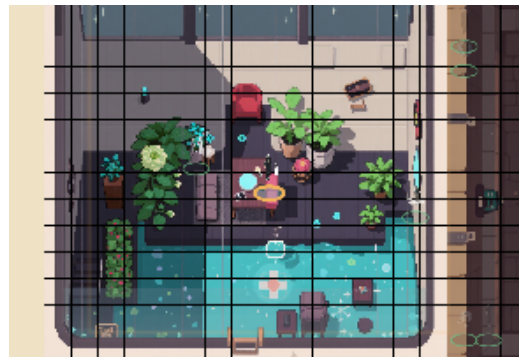
A Selected agent**B Agents in the rendered world****C Directional atlas**

Figure S7. Character identity in Tidal Embassy. (A) A crop of the interaction capture shows the selected character, marked by an amber ring. (B) A complementary world view situates agents among generated room surfaces and furniture. (C) The directional atlas preserves clothing, palette, and silhouette across movement. The interactive trial includes movement, messaging, item use, and trade.

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